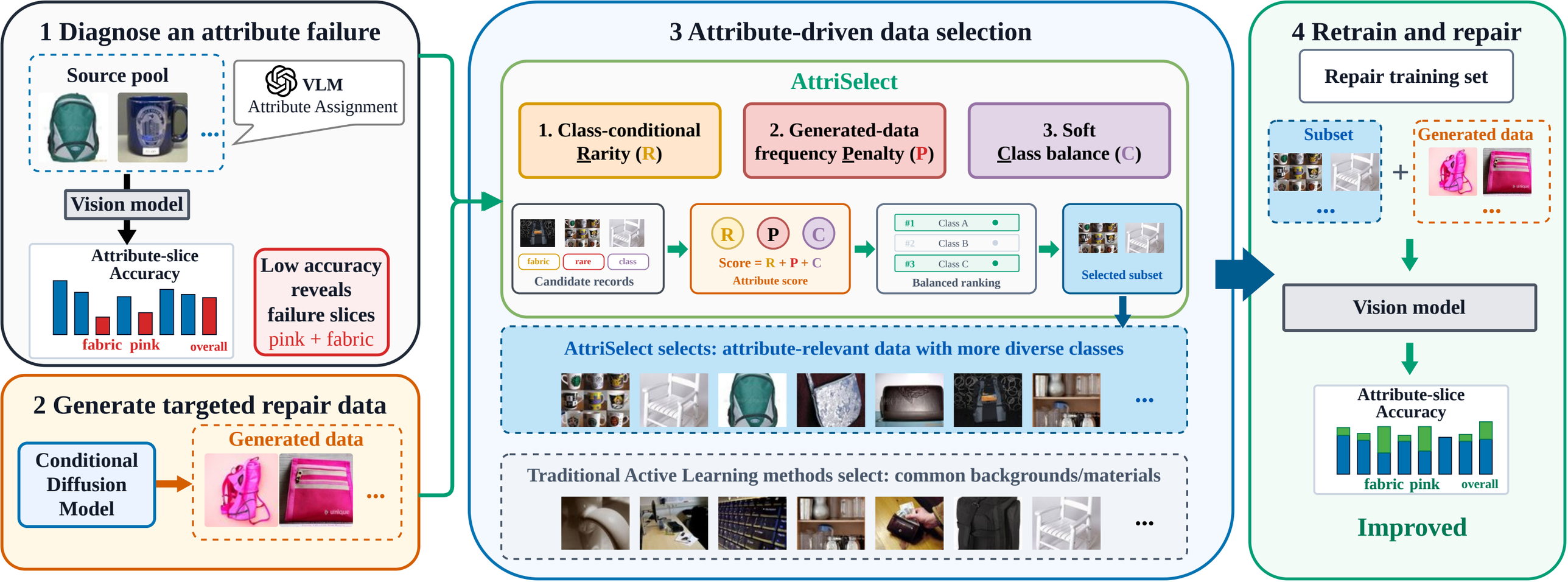

1 Diagnose an attribute failure
Source pool
...
VLM
Attribute Assignment
Vision model
Attribute-slice Accuracy
fabric
pink
overall
Low accuracy reveals failure slices
pink + fabric
2 Generate targeted repair data
Conditional Diffusion Model
Generated data
...
3 Attribute-driven data selection
AttriSelect
1. Class-conditional Rarity (R)
2. Generated-data frequency Penalty (P)
3. Soft Class balance (C)
fabric
rare
class
Candidate records
R
P
C
Score = R + P + C
Attribute score
#1 Class A
#2 Class B
#3 Class C
Balanced ranking
Selected subset
AttriSelect selects: attribute-relevant data with more diverse classes
...
Traditional Active Learning methods select: common backgrounds/materials
...
4 Retrain and repair
Repair training set
Subset
...
+
Generated data
...
Vision model
Attribute-slice Accuracy
fabric
pink
overall
Improved